



Qualification Pack

Machine Operator & Programmer CNC Lathe

QP Code: CPC/CAP/Q0701

Version: 1.0

NSQF Level: 4

CENTRAL INSTITUTE OF PETROCHEMICALS ENGINEERING & TECHNOLOGY || CIPET Head Office, T.V.K.
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CPC/CAP/Q0701: Machine Operator & Programmer CNC Lathe

Brief Job Description

The Machine operator & Programmer handle the CNC Lathe Machine, able to generate the program & operate the machine to make precision jobs

Personal Attributes

Technical aptitude, Safety-conscious, Reliability and punctuality, Organizational skills, Problem-solving ability, Teamwork and communication, Quality-focused mindset, Willingness to learn

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CPC/CAP/N0701: Basic Engineering](#)
2. [CPC/CAP/N0702: Metal Cutting & Cutting Tools](#)
3. [CPC/CAP/N0703: Conventional Lathe Machines](#)
4. [CPC/CAP/N0704: CNC Lathe Machine Programming](#)
5. [CPC/CAP/N0705: CNC Lathe Machine operation](#)
6. [CPC/CAP/N0406: To Work effectively with Others](#)
7. [CPC/CAP/N0408: Basics of computer and data entry](#)
8. [CPC/CAP/N0407: Soft Skills.](#)
9. [CPC/CAP/N0401: Basic health and safety practices at the workplace, 5S](#)
10. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Chemical & PetroChemical
Sub-Sector	Petrochemicals
Occupation	CNC Lathe, CNC Lathe

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Country	India
NSQF Level	4
Credits	22
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7223.6001
Minimum Educational Qualification & Experience	12th Class (or Equivalent) OR 10th Class (or Equivalent) with 2 Years of experience NTC/NAC OR 11th Class (or Equivalent) with 1.5 years of experience Relevance Experience OR 10th Class (or Equivalent) with 3 Years of experience Relevant Experience OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience Relevant Experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	25/05/2028
NSQC Approval Date	26/05/2025
Version	1.0
Reference code on NQR	QG-04-CP-04126-2025-V2-CIPET
NQR Version	2

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CPC/CAP/N0701: Basic Engineering

Description

This NOS is machine tool used for shaping or machining parts by rotating a workpiece against a cutting tool. It is a highly precise and automated tool commonly used in manufacturing processes.

Scope

The scope covers the following :

- This NOS is about developing the fundamental engineering knowledge and practical skills necessary for operating, maintaining, and assisting in basic engineering activities in a plastics manufacturing or processing environment

Elements and Performance Criteria

Apply Basic Engineering Concepts

To be competent, the user/individual on the job must be able to:

- PC1.** Understanding of basics of bench working
- PC2.** Identify various hand tools
- PC3.** Importance of filling & fitting
- PC4.** Understanding various tools used for filling & fitting operation
- PC5.** Follow the Procedure of filling

Perform Basic Maintenance and Housekeeping

To be competent, the user/individual on the job must be able to:

- PC6.** Able to make various fittings like V fit, square fit, dovetail fit
- PC7.** Able to operate the measuring instruments
- PC8.** Measurement of outside diameter & inside diameter of shaft without any error
- PC9.** Able to measure pitch of threads and taper angles

Use Basic Measuring Instruments

To be competent, the user/individual on the job must be able to:

- PC10.** Do simple calculations to find the weight, angles & tapers
- PC11.** Understanding of basics of drawing.
- PC12.** Able to make isometric and orthographic projections
- PC13.** Understanding drawings for preparation of process planning
- PC14.** Understanding and interpretation of geometric tolerances
- PC15.** Able to make & measure the job precisely

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Units of measurement and their conversions (metric and imperial)
- KU2.** Basic concepts of mechanics force, work, power, torque, and motion
- KU3.** Principles of electricity voltage, current, resistance, grounding, and circuit safety
- KU4.** Basic hydraulics and pneumatics components, symbols, and functions
- KU5.** Common machine elements bearings, gears, belts, pulleys, chains, couplings, fasteners, seals, and gaskets.
- KU6.** Types and functions of tools, tackles, and instruments used in maintenance and fabrication
- KU7.** Purpose and correct use of measuring instruments such as vernier calipers, micrometers, gauges, and thermometers
- KU8.** Methods of calibration and importance of measurement accuracy
- KU9.** Interpretation of dimensional tolerances and basic engineering drawings
- KU10.** Recording and reporting of measurement data in standard formats
- KU11.** Routine preventive maintenance practices lubrication, tightening, alignment, and cleaning
- KU12.** Basic troubleshooting techniques for mechanical and electrical faults
- KU13.** Identification of abnormal sounds, vibration, leakage, or overheating in machines
- KU14.** Safe use and storage of hand tools, lubricants, and cleaning materials
- KU15.** Importance of maintaining machine history cards and maintenance schedules
- KU16.** Use and importance of Personal Protective Equipment (PPE)
- KU17.** Potential hazards in engineering work (mechanical, electrical, chemical, thermal) and ways to minimize them
- KU18.** Safe handling of materials, tools, and equipment to prevent injury or damage
- KU19.** Procedures for safe shutdown and isolation of machines (Lockout/Tagout)
- KU20.** Waste management procedures segregation, disposal, and recycling of scrap and oil
- KU21.** Environmental protection norms and the importance of energy conservation
- KU22.** Methods of effective workplace communication and coordination.
- KU23.** Reporting formats for faults, maintenance completion, and inspection findings
- KU24.** Importance of timely and accurate documentation for quality and audit purposes
- KU25.** Use of simple computer applications (if applicable) for data entry, reporting, and communication

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret basic engineering drawings, labels, signs, process charts, and safety instructions
- GS2.** Read and understand technical manuals, maintenance schedules, and standard operating procedures
- GS3.** Write simple reports, maintenance logs, and inspection records clearly and legibly
- GS4.** Communicate effectively with supervisors, maintenance staff, and team members using appropriate terminology

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- GS5.** Ask questions to clarify instructions, resolve doubts, and seek guidance when necessary
- GS6.** Avoid use of jargon, slang, or acronyms when communicating unless commonly used in the workplace
- GS7.** Participate effectively in team briefings, safety meetings, and problem-solving discussions
- GS8.** Perform basic arithmetic operations addition, subtraction, multiplication, and division
- GS9.** Apply basic geometry and trigonometry for measurements, angles, and alignments
- GS10.** Convert units of measurement (mm to cm, bar to psi, etc.) as required in engineering calculations
- GS11.** Estimate material requirements, quantities, and time based on work instructions
- GS12.** Analyze simple numerical data from gauges, meters, and machine readings to detect abnormalities
- GS13.** Apply basic problem-solving and troubleshooting skills for common operational issues
- GS14.** Fill up inspection sheets, maintenance checklists, and machine logbooks in standard company format
- GS15.** Use digital devices (tablets, computers) to record data or report faults if applicable
- GS16.** Understand basic digital tools used for inventory, documentation, and process monitoring
- GS17.** Follow data entry protocols to maintain accuracy and confidentiality of information
- GS18.** Work effectively as part of a team, respecting roles and responsibilities of others
- GS19.** Support colleagues in achieving production or maintenance goals
- GS20.** Adapt to different roles and multitask as per job requirements
- GS21.** Learn continuously by observing, seeking feedback, and applying new knowledge
- GS22.** Share ideas and suggestions for process improvement and workplace efficiency
- GS23.** Demonstrate punctuality, discipline, and responsibility at the workplace
- GS24.** Follow company rules, ethical practices, and code of conduct
- GS25.** Take ownership for quality and safety of own work
- GS26.** Maintain a clean and organized work environment following 5S principles
- GS27.** Identify causes of potential delays or issues and take preventive actions
- GS28.** Handle tools and machinery with care to avoid damage or wastage
- GS29.** Stay calm and act objectively during emergencies or stressful situations
- GS30.** Show willingness to upgrade technical skills through training or practice

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Apply Basic Engineering Concepts</i>	7	15	-	-
PC1. Understanding of basics of bench working	1	3	-	-
PC2. Identify various hand tools	1	4	-	-
PC3. Importance of filling & fitting	1	3	-	-
PC4. Understanding various tools used for filling & fitting operation	2	2	-	-
PC5. Follow the Procedure of filling	2	3	-	-
<i>Perform Basic Maintenance and Housekeeping</i>	9	16	-	-
PC6. Able to make various fittings like V fit, square fit, dovetail fit	3	4	-	-
PC7. Able to operate the measuring instruments	2	4	-	-
PC8. Measurement of outside diameter & inside diameter of shaft without any error	2	4	-	-
PC9. Able to measure pitch of threads and taper angles	2	4	-	-
<i>Use Basic Measuring Instruments</i>	14	19	-	-
PC10. Do simple calculations to find the weight, angles & tapers	3	4	-	-
PC11. Understanding of basics of drawing.	2	3	-	-
PC12. Able to make isometric and orthographic projections	3	3	-	-
PC13. Understanding drawings for preparation of process planning	2	4	-	-
PC14. Understanding and interpretation of geometric tolerances	2	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Able to make & measure the job precisely	2	3	-	-
NOS Total	30	50	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0701
NOS Name	Basic Engineering
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	CNC Lathe, CNC Lathe
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0702: Metal Cutting & Cutting Tools

Description

This NOS is designed for candidates in manufacturing, toolroom, or plastic processing environments where basic machining, metal preparation, and maintenance are essential.

Scope

The scope covers the following :

- This NOS is about developing the fundamental engineering knowledge and practical skills necessary for operating, maintaining, and assisting in basic engineering activities in a plastics manufacturing or processing environment.

Elements and Performance Criteria

Identify and Understand Cutting Tools

To be competent, the user/individual on the job must be able to:

PC1. Discuss about the type of materials/metals being used in the industry & for making jobs

PC2. • Refer all tool material to understand properties and able to select the tool according to the job requirements

PC3. • Able to understand various cutting mechanisms of various machines like Lathe, milling, drilling, Grinding

Understand Metal Cutting Principles

To be competent, the user/individual on the job must be able to:

PC4. Selecting proper tool according to the job requirement

PC5. Ensure that the proper tool is available before initiating the job

Handle and Maintain Cutting Tools

To be competent, the user/individual on the job must be able to:

PC6. Understanding of various cutting fluids used in manufacturing

PC7. Able to understand the application of various tools

PC8. Understanding of various cutting parameters and selecting proper parameters for the job

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Organization manufacturing processes and types of machines used

KU2. Safety policies, code of conduct, and 5S practices followed on the shop floor

KU3. SOPs for machining, tool handling, and preventive maintenance

KU4. Quality parameters for machined components (surface finish, tolerance, accuracy)

KU5. Importance of productivity, cost control, and material conservation

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- KU6.** Documentation procedures for tool issue, return, and maintenance logs
- KU7.** Escalation procedures for machine faults or safety hazards
- KU8.** Types of metals and their machinability characteristics
- KU9.** Basic concepts of metal cutting speed, feed, depth of cut, tool life
- KU10.** Tool materials, coatings, and their selection criteria
- KU11.** Cutting tool geometry and its effect on chip formation and surface finish
- KU12.** Functions and use of common tools: lathe tool, drill, reamer, tap, end mill, saw, and grinder
- KU13.** Tool holding devices (collet, chuck, arbour, fixture, magnetic table, etc.)
- KU14.** Common machining defects and methods to prevent them
- KU15.** Principles of tool sharpening and maintenance
- KU16.** Role and properties of cutting fluids cooling, lubrication, chip evacuation
- KU17.** Standard safety symbols and signage used in engineering workshops
- KU18.** Procedure for tool calibration, inspection, and life tracking
- KU19.** Procedure for tool calibration, inspection, and life tracking

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read work instructions, drawings, and operation sheets
- GS2.** Write brief notes or reports regarding tool condition, breakage, or maintenance
- GS3.** Communicate effectively with team members and supervisors about machining tasks
- GS4.** Follow verbal and written instructions accurately
- GS5.** Participate in safety and quality meetings
- GS6.** Perform simple calculations involving speed, feed, and depth of cut
- GS7.** Use measuring instruments such as scale, vernier caliper, micrometer, and gauge blocks
- GS8.** Interpret readings from machine dials, meters, and gauges
- GS9.** Identify tool wear patterns and analyze probable causes
- GS10.** Apply basic geometry in determining tool angles and cutting parameters
- GS11.** Identify minor faults and suggest corrective actions
- GS12.** Identify minor faults and suggest corrective actions
- GS13.** Plan and prioritize work as per production schedule
- GS14.** Support continuous improvement and 5S activities
- GS15.** Demonstrate discipline, punctuality, and responsibility at the workplace
- GS16.** Follow safe lifting, clamping, and handling practices
- GS17.** Maintain a clean and organized workstation
- GS18.** Dispose of metal scrap and coolant as per environmental norms
- GS19.** Respond to emergency situations calmly and promptly
- GS20.** Continuously monitor and control noise, heat, and dust levels at the workstation

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify and Understand Cutting Tools</i>	7	7	-	-
PC1. Discuss about the type of materials/metals being used in the industry & for making jobs	2	3	-	-
PC2. • Refer all tool material to understand properties and able to select the tool according to the job • requirements	3	2	-	-
PC3. • Able to understand various cutting mechanisms of various machines like Lathe, milling, drilling, • Grinding	2	2	-	-
<i>Understand Metal Cutting Principles</i>	4	5	-	-
PC4. Selecting proper tool according to the job requirement	2	2	-	-
PC5. Ensure that the proper tool is available before initiating the job	2	3	-	-
<i>Handle and Maintain Cutting Tools</i>	9	8	-	-
PC6. Understanding of various cutting fluids used in manufacturing	3	3	-	-
PC7. Able to understand the application of various tools	3	2	-	-
PC8. Understanding of various cutting parameters and selecting proper parameters for the job	3	3	-	-
NOS Total	20	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0702
NOS Name	Metal Cutting & Cutting Tools
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	CNC Lathe, CNC Lathe
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

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CPC/CAP/N0703: Conventional Lathe Machines

Description

This NOS is meant for trainees or operators learning operation, setup, and basic maintenance of conventional (manual) lathe machines a key foundational unit in mechanical and plastics tooling courses

Scope

The scope covers the following :

- This NOS covers the competencies required to set up, operate, and maintain conventional lathe machines for turning, facing, drilling, and finishing of metallic and non-metallic components, while ensuring accuracy, safety, and quality

Elements and Performance Criteria

Identify and Understand Lathe Machine Components To be competent, the individual must be able to

To be competent, the user/individual on the job must be able to:

- PC1.** Understand and comply with safety, environmental & other relevant regulations and guidelines
- PC2.**
 - Wear personal protective equipment (PPE) like safety glasses, apron, no loose cloths/ hair, safety shoes while performing lathe operations regulations while performing CNC turning operations
- PC3.** Ensure work area is clean and safe
- PC4.** Ensure that machine safety guards are in place and are in correctly working condition
- PC5.** Ensure that all tools, equipment are in safe and usable conditions
- PC6.** Ensure availability of job specification i.e. approved drawings, sketches, instructions from the supervisor, job instruction sheet/ job card

Prepare the Lathe and Workpiece for Operation To be competent, the individual must be able to

To be competent, the user/individual on the job must be able to:

- PC7.**
 - Read and understand the Job requirements from the job specifications and attention shall be given to the geometric tolerances
- PC8.**
 - Check the work piece material for the dimensions and ensure that it is free from foreign objects, dirt or other contamination and is within the required size
- PC9.**
 - Plan to perform the turning or other lathe operations and the sequence of operations as per required job specifications
- PC10.** Obtain all the appropriate tools and measuring instruments/ gauges required for the job
- PC11.** Check the lathe machine for its functioning and ensure that it is ready for operation
- PC12.**
 - Prepare the lathe machine for the operations by mounting and setting the required work holding devices and cutting tools

Perform Basic Lathe Operations

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To be competent, the user/individual on the job must be able to:

- PC13.** • Clarify any doubt, if any and see necessary instruction /training on the operation of the machine
 - whenever required
- PC14.** Hold the work piece securely and correctly, without distortion
- PC15.** Adjust the machine settings as per job requirement to maintain desired accuracy
- PC16.** Adjust and set the speed and feed of the lathe machine to achieve the job specifications
- PC17.** • Operate the machine tool controls safely and correctly, in line with operational procedures both
 - in manual and power modes
- PC18.** • Stop the lathe machine, both in normal and emergency situations correctly by following the right
 - procedure and should be able to restart the machine after and emergency

Maintain and Care for Lathe Tools and Equipment

To be competent, the user/individual on the job must be able to:

- PC19.** • Should be able to use the lathe machine accessories and attachments such as steady and follower
 - rests, tail stock, taper turning attachments, profile attachments etc
- PC20.** • Perform various lathe operations using different tools to produce components with various
 - features
- PC21.** Produce components as per required quality standards and free from burrs & sharp edges
- PC22.** Shall achieve given production targets
- PC23.** • Shall be able to apply roughing and finishing cuts, considering the effect on tool life, surface finish
 - and dimensional accuracy
- PC24.** • Shall be able to use coolants/ cutting fluids for different combinations of work piece and tool as per
 - different locations

Follow Health, Safety, and Environmental Practices

To be competent, the user/individual on the job must be able to:

- PC25.** • Shall be able to observe and report any difficulties/ discrepancies that may arise during the machine
 - operation and carry out the corrective actions as per instructions
- PC26.** • Correctly shutting down the machine on completion of the machining operations, removing and
 - disposing of the chips/ waste and critical parameters different locations
- PC27.** Use of measuring instruments/ gauges to check the critical parameters
- PC28.** Shall be able to carry out the corrective action, in the case of deviation from the required specifications
- PC29.** Report the problem to the supervisor, if it cannot be resolved
- PC30.** • Seek guidance from the supervisor/ specialist of the problem is outside his/her area of
 - competence

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Policies and procedures for machine operation, inspection, and maintenance
- KU2.** Organizational structure and reporting hierarchy
- KU3.** Importance of productivity, accuracy, and material conservation
- KU4.** Company quality control standards for machining
- KU5.** housekeeping practices in the workshop
- KU6.** Documentation formats job cards, tool logs, and inspection records
- KU7.** SOPs for emergency shutdowns and workplace safety
- KU8.** Types and functions of conventional lathes engine, bench, and capstan lathes
- KU9.** Lathe specifications and terminology (bed length, swing, spindle speed range)
- KU10.** Basic cutting principles speed, feed, depth of cut, and material removal rate
- KU11.** Types of lathe tools and tool angles for various operations
- KU12.** Selection of cutting tools and materials based on workpiece material
- KU13.** Lathe attachments taper turning attachment, thread chasing dial, tool post grinder
- KU14.** Measuring instruments and their proper use for accuracy verification
- KU15.** Common lathe defects (chatter, poor finish, taper errors) and corrective measures
- KU16.** Lubricants and coolants used in turning operations
- KU17.** Safe handling and storage of tools, chucks, and accessories
- KU18.** Preventive maintenance and machine lubrication schedule
- KU19.** Importance of balancing, alignment, and vibration control
- KU20.** Environmental and energy-saving practices in machining

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read job instructions, drawings, and process sheets
- GS2.** Write inspection readings, tool conditions, and maintenance remarks
- GS3.** Communicate effectively with co-workers and supervisors
- GS4.** Understand and follow verbal instructions accurately
- GS5.** Participate in shift handover communication effectively
- GS6.** Perform calculations related to cutting speed, feed, and time
- GS7.** Use measuring tools (vernier, micrometer, thread gauge) for inspection
- GS8.** Estimate material removal and tool wear
- GS9.** Analyze reasons for dimensional errors or poor surface finish
- GS10.** Apply simple geometry and trigonometry in taper and thread cutting
- GS11.** Work collaboratively with maintenance and quality teams
- GS12.** Share tools and workspace responsibly with peers
- GS13.** Take initiative to learn advanced lathe techniques (thread cutting, knurling)
- GS14.** Adapt to different lathe operations and job requirements
- GS15.** Seek feedback and improve skills continuously



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- GS16.** Maintain cleanliness and order around the machine
- GS17.** Follow safe chip removal and tool handling practices
- GS18.** Dispose of waste materials as per environmental norms
- GS19.** Report unsafe conditions immediately
- GS20.** Stay calm and act responsibly in emergencies (machine jam, fire, injury)

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify and Understand Lathe Machine Components To be competent, the individual must be able to</i>	6	8	-	-
PC1. Understand and comply with safety, environmental & other relevant regulations and guidelines	1	2	-	-
PC2. • Wear personal protective equipment (PPE) like safety glasses, apron, no loose cloths/ hair, safety shoes while performing lathe operations regulations while performing CNC turning operations	1	2	-	-
PC3. Ensure work area is clean and safe	1	1	-	-
PC4. Ensure that machine safety guards are in place and are in correctly working condition	1	1	-	-
PC5. Ensure that all tools, equipment are in safe and usable conditions	1	1	-	-
PC6. Ensure availability of job specification i.e. approved drawings, sketches, instructions from the supervisor, job instruction sheet/ job card	1	1	-	-
<i>Prepare the Lathe and Workpiece for Operation To be competent, the individual must be able to</i>	6	12	-	-
PC7. • Read and understand the Job requirements from the job specifications and attention shall be given to the geometric tolerances	1	1	-	-
PC8. • Check the work piece material for the dimensions and ensure that it is free from foreign objects, dirt or other contamination and is within the required size	1	1	-	-
PC9. • Plan to perform the turning or other lathe operations and the sequence of operations as per required job specifications	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Obtain all the appropriate tools and measuring instruments/ gauges required for the job	1	2	-	-
PC11. Check the lathe machine for its functioning and ensure that it is ready for operation	1	3	-	-
PC12. • Prepare the lathe machine for the operations by mounting and setting the required work holding devices and cutting tools	1	3	-	-
<i>Perform Basic Lathe Operations</i>	6	16	-	-
PC13. • Clarify any doubt, if any and see necessary instruction /training on the operation of the machine • whenever required	1	3	-	-
PC14. Hold the work piece securely and correctly, without distortion	1	3	-	-
PC15. Adjust the machine settings as per job requirement to maintain desired accuracy	1	3	-	-
PC16. Adjust and set the speed and feed of the lathe machine to achieve the job specifications	1	3	-	-
PC17. • Operate the machine tool controls safely and correctly, in line with operational procedures both in manual and power modes	1	2	-	-
PC18. • Stop the lathe machine, both in normal and emergency situations correctly by following the right procedure and should be able to restart the machine after and emergency	1	2	-	-
<i>Maintain and Care for Lathe Tools and Equipment</i>	6	12	-	-
PC19. • Should be able to use the lathe machine accessories and attachments such as steady and follower • rests, tail stock, taper turning attachments, profile attachments etc	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC20. Perform various lathe operations using different tools to produce components with various features	1	2	-	-
PC21. Produce components as per required quality standards and free from burrs & sharp edges	1	2	-	-
PC22. Shall achieve given production targets	1	2	-	-
PC23. Shall be able to apply roughing and finishing cuts, considering the effect on tool life, surface finish and dimensional accuracy	1	2	-	-
PC24. Shall be able to use coolants/ cutting fluids for different combinations of work piece and tool as per different locations	1	2	-	-
<i>Follow Health, Safety, and Environmental Practices</i>	6	12	-	-
PC25. Shall be able to observe and report any difficulties/ discrepancies that may arise during the machine operation and carry out the corrective actions as per instructions	1	2	-	-
PC26. Correctly shutting down the machine on completion of the machining operations, removing and disposing of the chips/ waste and critical parameters different locations	1	2	-	-
PC27. Use of measuring instruments/ gauges to check the critical parameters	1	2	-	-
PC28. Shall be able to carry out the corrective action, in the case of deviation from the required specifications	1	2	-	-
PC29. Report the problem to the supervisor, if it cannot be resolved	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC30. Seek guidance from the supervisor/ specialist of the problem is outside his/her area of competence	1	2	-	-
NOS Total	30	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0703
NOS Name	Conventional Lathe Machines
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	CNC Lathe, CNC Lathe
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0704: CNC Lathe Machine Programming

Description

This NOS is designed for learners who will write, test, and optimize CNC lathe programs and operate CNC turning centers in line with industrial standards

Scope

The scope covers the following :

- This NOS covers the skills, knowledge, and understanding required to create, verify, and execute CNC lathe programs for precision machining of metal and non-metal components, ensuring accuracy, efficiency, and adherence to safety and quality standards

Elements and Performance Criteria

Understand CNC Lathe System and Components To be competent, the individual must be able to

To be competent, the user/individual on the job must be able to:

- PC1.** Comply with health and safety, environmental and other relevant regulations at work
- PC2.** Adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while programming CNC Lathe machines
- PC3.** Work following laid down procedures and instructions
- PC4.** Ensure that machine guards are in place and are correctly adjusted
- PC5.** Read and understand safety instructions, warning signs on the machine
- PC6.** Ensure work area is clean and safe from hazards
- PC7.**
 - Ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition
- PC8.**
 - Ensure availability of job specification i.e. approved drawings, sketches, instructions from the supervisor, job instruction sheet/ job card
- PC9.** Read and establish job requirements from the job specification document accurately
- PC10.** Follow job instructions, assembly drawings and laid down procedures at all times

Prepare Work Setup for CNC Turning To be competent, the individual must be able to

To be competent, the user/individual on the job must be able to:

- PC11.**
 - Report and rectify incorrect and inconsistent information in job specification documents as per organization procedures
- PC12.** Use and extract information from reference charts, tables, graphs and standards
- PC13.** Prepare the work area as per procedure or operational specification
- PC14.** Conduct a preliminary check of the readiness of the program so that the CNC machine operates correctly
- PC15.**
 - Determine what operational objectives and targets need to be achieved and how best the machine needs to be programmed to achieve this

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- PC16.** Extract and use information from engineering drawings and related specifications in relation to work undertaken
- PC17.** Identify tool requirements from tooling layout and assess their suitability
- PC18.** Identify suitable work holding or fix turing device as per the job requirement
- PC19.** Ensure the correct and latest part-program is uploaded onto the CNC system
- PC20.** Make CNC program with commands for tool motions, spindle motions, misc. functions & tool change, in syntax corresponding to machine and control system on which the component will be machined

Write and Edit CNC Lathe Programs

To be competent, the user/individual on the job must be able to:

- PC21.** Various ways to make CNC program are by writing it on paper or in computer's text editor, or using CAM software or controllers on machine Ways: written, directly entered into machine controller, using computer software- CAM software
- PC22.** Ensure that part program is efficient & results in minimal cycle time, with optimal cutting parameters and no unnecessary tool motions
- PC23.** Use subprograms and canned cycles, to reduce program size and input time and avoid memory overflow on the machine
- PC24.**
 - Transfer the program to the machine by entering it at the console or transmitting it through a
 - wired link or through a data transfer device
- PC25.** Follow the correct procedures for calling up the program and dealing with any error messages or faults
- PC26.**
 - Handle the typical problems that can occur with the programming, loading and editing activities
 - effectively using approved procedures
- PC27.**
 - Save the proven program in the appropriate storage medium – paper, computer hard disk, etc.
 - and
 - location
- PC28.** Complete relevant documentation as per organizational procedure
- PC29.** Leave the work area in a safe and tidy condition on completion of the activities
- PC30.** Obtain appropriate equipment or tools needed as per job requirements

Simulate and Execute CNC Programs

To be competent, the user/individual on the job must be able to:

- PC31.** Ensure that all measuring equipment is calibrated and approved for usage
- PC32.**
 - Ensure that the tools and fixtures are in usable condition (Eg. free from breakage, damage,
 - calibration, etc.)
- PC33.** Pre-set tooling appropriately using setting jigs/ fixtures
- PC34.** Seek any necessary instruction/training on operation of the machine where required
- PC35.** Mount tools in the correct positions in the tool turret or magazine
- PC36.** Check that the tools have been mounted in positions corresponding to tool numbers in the part program
- PC37.** Mount the part on machine firmly in the specified work holding devices, with the appropriate clamping forces
- PC38.** Enter work offset and tool data on the machine – X and Z offsets, tool orientation and radius for lathes; length offsets and tool radius for machining centers

Qualification Pack

PC39. Ensure that tool data has been entered in offset number corresponding to tool offset numbers in part program

PC40. Deal with error messages and faults on the program or equipment

Maintain Quality, Documentation, and Safety To be competent, the individual must be able to

To be competent, the user/individual on the job must be able to:

PC41. Cut a trial part using single block run, dry run and feed and speed override controls

PC42. Edit the program and adjust tool and wear offsets to correct any dimensional errors on the part

PC43. • Ensure that trial part conforms to drawing specifications in terms of dimensions, surface finishes &
• geometrical parameters like concentricity, parallelism, run out, etc

PC44. Correct the tool wear offsets whenever required, based on the results of the period inspection

PC45. Change worn out tools and indexable inserts whenever required

PC46. After every change of a worn out tool or insert, cut a trial part and correct any dimensional inaccuracies by adjusting the tool offsets or wear offsets

PC47. Return worn out cutting tools, work holding device / fixtures / instruments / drawings to store

PC48. Ensure that there is no damage to the tool/fixture while doing the prove-out

PC49. Shut down the equipment to a safe condition on conclusion of the activities

PC50. • Deal promptly and effectively with problems within span of responsibility and control and report
• those that cannot be solved

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Organizational policies and SOPs for CNC programming, inspection, and maintenance

KU2. Job reporting hierarchy and communication protocols

KU3. Documentation formats for job setup, offset data, and tool life monitoring

KU4. Quality management system (ISO 9001 / IATF 16949) procedures related to machining

KU5. Importance of productivity, accuracy, and safety in CNC operations

KU6. Organizational procedures for emergency handling, fire safety, and machine breakdown

KU7. Principles and working of CNC lathe machines (2-axis, 3-axis)

KU8. Functions and operation of CNC control systems (Fanuc, Siemens, Haas, etc.)

KU9. Machine coordinate system and tool offset principles

KU10. G-codes and M-codes used in CNC lathe programming

KU11. Canned cycles (G71, G72, G73, G74, G76, etc.) and their applications

KU12. Cutting parameters speed, feed, and depth of cut and their effect on tool life and finish

KU13. Tool nose radius compensation and coordinate system setting

KU14. Steps in program verification, simulation, and debugging

KU15. CNC data transfer methods (USB, DNC, Ethernet)

KU16. Preventive maintenance and lubrication of CNC machines

Qualification Pack

- KU17.** Measuring instruments for inspection micrometer, CMM, profile projector
- KU18.** Tool management systems and tool pre-setters
- KU19.** Basics of CAD/CAM integration for CNC programming
- KU20.** Safety practices and interlocks in CNC machines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret engineering drawings, part programs, and operation manuals
- GS2.** Record machining data, inspection results, and maintenance details
- GS3.** Communicate effectively with team members and supervisors
- GS4.** Use technical terms and codes accurately while writing programs
- GS5.** Report deviations or machine alarms clearly and promptly
- GS6.** Calculate spindle speed, feed rate, and cutting time based on material and tool data
- GS7.** Apply geometry and trigonometry to coordinate calculations and part features
- GS8.** Analyze tool wear and cycle time to optimize performance.
- GS9.** Interpret measurement data to verify part conformance
- GS10.** Use logical reasoning for program debugging and error correction
- GS11.** Work collaboratively with quality, maintenance, and design teams
- GS12.** Adapt to different CNC controllers and software environments
- GS13.** Learn new machining cycles and programming methods proactively
- GS14.** Share feedback for improving setup efficiency and tool performance
- GS15.** Support peers during program trials or production runs
- GS16.** Follow PPE requirements (safety shoes, goggles, ear protection)
- GS17.** Keep the work area organized and free from obstructions
- GS18.** Handle tools, inserts, and measuring instruments safely
- GS19.** Dispose of coolant, chips, and lubricants per environmental standards
- GS20.** Follow machine lockout/tagout procedures during maintenance or setup changes

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand CNC Lathe System and Components To be competent, the individual must be able to</i>	5	11	-	-
PC1. Comply with health and safety, environmental and other relevant regulations at work	0.5	1	-	-
PC2. Adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while programming CNC Lathe machines	0.5	1	-	-
PC3. Work following laid down procedures and instructions	0.5	1	-	-
PC4. Ensure that machine guards are in place and are correctly adjusted	0.5	1	-	-
PC5. Read and understand safety instructions, warning signs on the machine	0.5	2	-	-
PC6. Ensure work area is clean and safe from hazards	0.5	1	-	-
PC7. • Ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable • condition	0.5	1	-	-
PC8. • Ensure availability of job specification i.e. approved drawings, sketches, instructions from the • supervisor, job instruction sheet/ job card	0.5	1	-	-
PC9. Read and establish job requirements from the job specification document accurately	0.5	1	-	-
PC10. Follow job instructions, assembly drawings and laid down procedures at all times	0.5	1	-	-
<i>Prepare Work Setup for CNC Turning To be competent, the individual must be able to</i>	5	10	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. • Report and rectify incorrect and inconsistent information in job specification documents as per • organization procedures	0.5	1	-	-
PC12. Use and extract information from reference charts, tables, graphs and standards	0.5	1	-	-
PC13. Prepare the work area as per procedure or operational specification	0.5	1	-	-
PC14. Conduct a preliminary check of the readiness of the program so that the CNC machine operates correctly	0.5	1	-	-
PC15. • Determine what operational objectives and targets need to be achieved and how best the machine • needs to be programmed to achieve this	0.5	1	-	-
PC16. Extract and use information from engineering drawings and related specifications in relation to work undertaken	0.5	1	-	-
PC17. Identify tool requirements from tooling layout and assess their suitability	0.5	1	-	-
PC18. Identify suitable work holding or fix turing device as per the job requirement	0.5	1	-	-
PC19. Ensure the correct and latest part-program is uploaded onto the CNC system	0.5	1	-	-
PC20. Make CNC program with commands for tool motions, spindle motions, misc. functions & tool change, in syntax corresponding to machine and control system on which the component will be machined	0.5	1	-	-
<i>Write and Edit CNC Lathe Programs</i>	7.5	19	-	-
PC21. Various ways to make CNC program are by writing it on paper or in computer's text editor, or using CAM software or controllers on machine Ways: written, directly entered into machine controller, using computer software- CAM software	0.5	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. Ensure that part program is efficient & results in minimal cycle time, with optimal cutting parameters and no unnecessary tool motions	0.5	2	-	-
PC23. Use subprograms and canned cycles, to reduce program size and input time and avoid memory overflow on the machine	0.5	2	-	-
PC24. • Transfer the program to the machine by entering it at the console or transmitting it through a • wired link or through a data transfer device	0.5	2	-	-
PC25. Follow the correct procedures for calling up the program and dealing with any error messages or faults	0.5	2	-	-
PC26. • Handle the typical problems that can occur with the programming, loading and editing activities • effectively using approved procedures	1	2	-	-
PC27. • Save the proven program in the appropriate storage medium – paper, computer hard disk, etc. and • location	1	2	-	-
PC28. Complete relevant documentation as per organizational procedure	1	2	-	-
PC29. Leave the work area in a safe and tidy condition on completion of the activities	1	2	-	-
PC30. Obtain appropriate equipment or tools needed as per job requirements	1	2	-	-
<i>Simulate and Execute CNC Programs</i>	7.5	20	-	-
PC31. Ensure that all measuring equipment is calibrated and approved for usage	0.5	2	-	-
PC32. • Ensure that the tools and fixtures are in usable condition (Eg. free from breakage, damage, • calibration, etc.)	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. Pre-set tooling appropriately using setting jigs/ fixtures	1	2	-	-
PC34. Seek any necessary instruction/training on operation of the machine where required	1	2	-	-
PC35. Mount tools in the correct positions in the tool turret or magazine	0.5	2	-	-
PC36. Check that the tools have been mounted in positions corresponding to tool numbers in the part program	1	2	-	-
PC37. Mount the part on machine firmly in the specified work holding devices, with the appropriate clamping forces	1	2	-	-
PC38. Enter work offset and tool data on the machine – X and Z offsets, tool orientation and radius for lathes; length offsets and tool radius for machining centers	0.5	2	-	-
PC39. Ensure that tool data has been entered in offset number corresponding to tool offset numbers in part program	0.5	2	-	-
PC40. Deal with error messages and faults on the program or equipment	0.5	2	-	-
<i>Maintain Quality, Documentation, and Safety To be competent, the individual must be able to</i>	5	20	-	-
PC41. Cut a trial part using single block run, dry run and feed and speed override controls	0.5	2	-	-
PC42. Edit the program and adjust tool and wear offsets to correct any dimensional errors on the part	0.5	2	-	-
PC43. - Ensure that trial part conforms to drawing specifications in terms of dimensions, surface finishes & - geometrical parameters like concentricity, parallelism, run out, etc	0.5	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC44. Correct the tool wear offsets whenever required, based on the results of the period inspection	0.5	2	-	-
PC45. Change worn out tools and indexable inserts whenever required	0.5	2	-	-
PC46. After every change of a worn out tool or insert, cut a trial part and correct any dimensional inaccuracies by adjusting the tool offsets or wear offsets	0.5	2	-	-
PC47. Return worn out cutting tools, work holding device / fixtures / instruments / drawings to store	0.5	2	-	-
PC48. Ensure that there is no damage to the tool/fixture while doing the prove-out	0.5	2	-	-
PC49. Shut down the equipment to a safe condition on conclusion of the activities	0.5	2	-	-
PC50. • Deal promptly and effectively with problems within span of responsibility and control and report • those that cannot be solved	0.5	2	-	-
NOS Total	30	80	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0704
NOS Name	CNC Lathe Machine Programming
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	CNC Lathe, CNC Lathe
NSQF Level	4
Credits	5
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQF Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0705: CNC Lathe Machine operation

Description

This NOS covers the competencies required to operate CNC lathe machines safely and efficiently to produce precision components as per job specifications

Scope

The scope covers the following :

- This NOS covers includes machine setup, operation, inspection, troubleshooting, and maintenance under minimal supervision

Elements and Performance Criteria

Prepare for CNC Lathe Operation

To be competent, the user/individual on the job must be able to:

- PC1.** Understand and comply with safety, environmental & other relevant regulations and guidelines
- PC2.** Wear personal protective equipment (PPE) like safety glasses, apron, no loose cloths/ hair, safety shoes while performing lathe operations while performing CNC turning operations
- PC3.** Ensure work area is clean and safe
- PC4.** Ensure that machine safety guards are in place and are in correctly working condition
- PC5.** Ensure that all tools, equipment are in safe and usable conditions
- PC6.** Ensure availability of job specification i.e. approved drawings, sketches, instructions from the supervisor, job instruction sheet/ job card
- PC7.**
 - Read and understand the Job requirements from the job specifications and attention shall be given
 - to the geometric tolerances
- PC8.** Check the work piece material for the dimensions and ensure that it is free from foreign objects, dirt or other contamination and is within the required size
- PC9.** Plan to perform the turning or other lathe operations and the sequence of operations as per required job specifications on CNC lathe machine
- PC10.** Obtain all the appropriate tools and measuring instruments/ gauges required for the job

Operate CNC Lathe Machine

To be competent, the user/individual on the job must be able to:

- PC11.** Check the CNC lathe machine for its functioning and ensure that it is ready for operation
- PC12.** Prepare the CNC lathe machine for the operations by mounting and setting the required work holding devices and cutting tools
- PC13.** Clarify any doubt, if any and see necessary instruction /training on the operation of the CNC Lathe machine whenever required
- PC14.** Hold the work piece securely and correctly, without distortion
- PC15.** Adjust the CNC Lathe machine settings as per job requirement to maintain desired accuracy

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- PC16.** Perform daily maintenance of machine according to defined checklist, at the beginning of day's shifts
- PC17.** Use and extract information from engineering drawings, dimensioning and tolerances
- PC18.** Use and extract information from reference charts, tables, graphs and Engineering standards
- PC19.** Load and unload component(s) using pre-determined fixtures or work holding devices as per work instructions
- PC20.** Make basic program and check correctness of program through dry run and single block check

Inspect and Measure Components

To be competent, the user/individual on the job must be able to:

- PC21.** Adjust and set the speed and feed of the CNC lathe machine to achieve the job specifications
- PC22.** Operate the machine tool controls safely and correctly, in line with operational procedures
- PC23.** Stop the CNC lathe machine, both in normal and emergency situations correctly by following the right procedure and should be able to restart the machine after the emergency
- PC24.** Do first part cutting trial by setting tool offsets to get oversize part
- PC25.**
 - Measure the critical parameters of the machined component on the machine (without removing from the machine)
- PC26.** Correct the offsets based on the measurements by accessing program edit facility in order to enter tooling data
- PC27.** Measure the component after unloading to check for accuracy in the critical parameters as per job specifications
- PC28.** Produce machined components that combine different turning operations and have a range of features
- PC29.** Follow the specified machining sequence and procedure as per job specifications
- PC30.** Interpret in-built machine alarms & respond to same as per operating manual or specified instructions

Maintain Machine and Workplace

To be competent, the user/individual on the job must be able to:

- PC31.** Observe for inconsistency in dimensions due to tool wear and correct the offsets accordingly
- PC32.** Ensure that machine settings are adjusted as and when required, either by self or the setter, to maintain the required accuracy
- PC33.** Identify when tools need replacement and replace worn tool with new tool
- PC34.** Produce components as per required standards
- PC35.** Report problems and seek appropriate assistance in a timely manner
- PC36.**
 - Complete documentation during and post operations as per organizational procedures and applicable quality management system
- PC37.** Return the machine and all tools and equipment to the correct location on completion of activities
- PC38.** Leave the work area in a safe and tidy condition on completion of job activities as per 5S practices
- PC39.** Report the problem to the supervisor, if it cannot be resolved

Qualification Pack

PC40. Seek guidance from the supervisor/ specialist of the problem is outside his/her area of competence

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company policies on quality, safety, and operational procedures
- KU2.** Roles and responsibilities within the production department
- KU3.** Formats for job instructions, inspection records, and maintenance logs
- KU4.** Reporting hierarchy and communication channels for technical issues
- KU5.** Quality management and documentation system followed in the organization
- KU6.** Working principle of CNC lathe machines and their components
- KU7.** Axis configuration, coordinate systems, and work offsets
- KU8.** Tooling types used in CNC turning (holders, inserts, boring bars, drills, etc.)
- KU9.** Common machining operations facing, turning, threading, grooving, drilling, parting
- KU10.** Importance of cutting parameters speed, feed, and depth of cut
- KU11.** Methods of tool setting and offset input
- KU12.** Types of workholding devices chuck, collet, mandrel, centers
- KU13.** Basics of G-codes and M-codes relevant to CNC lathe operation
- KU14.** Common machine alarms and troubleshooting methods
- KU15.** Measuring instruments and their usage for dimensional inspection
- KU16.** Importance of maintaining tool life and surface finish
- KU17.** Coolant systems and chip management
- KU18.** Safety features of CNC lathe (interlocks, E-stop, door switch)
- KU19.** Preventive maintenance requirements
- KU20.** Environmental and occupational health safety guidelines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret engineering drawings and work instructions
- GS2.** Write clear and concise inspection and production reports
- GS3.** Communicate effectively with supervisors and maintenance teams
- GS4.** Follow verbal and written instructions accurately
- GS5.** Report operational issues promptly and clearly
- GS6.** Perform basic arithmetic calculations (speed, feed, depth of cut, machining time)
- GS7.** Use geometry and trigonometry for interpreting part profiles and tool paths
- GS8.** Estimate tool wear and dimensional variation through analytical reasoning
- GS9.** Analyze data from inspection reports for corrective actions

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- GS10.** Work cooperatively with co-workers to achieve production targets
- GS11.** Learn new machining cycles and operation methods through observation and feedback
- GS12.** Support team members in setup, changeover, and troubleshooting
- GS13.** Share suggestions for process and quality improvement
- GS14.** Demonstrate responsibility for assigned tasks and outcomes
- GS15.** Use PPEs such as gloves, safety shoes, goggles, and ear protection
- GS16.** Maintain proper posture and ergonomics while operating machines
- GS17.** Ensure the machine door and guards are closed before cycle start
- GS18.** Follow fire and electrical safety procedures
- GS19.** Handle tools, coolant, and chips safely and dispose of waste responsibly
- GS20.** Respond appropriately to emergencies, alarms, or equipment malfunctions

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for CNC Lathe Operation</i>	6	16	-	-
PC1. Understand and comply with safety, environmental & other relevant regulations and guidelines	0.5	1	-	-
PC2. Wear personal protective equipment (PPE) like safety glasses, apron, no loose cloths/ hair, safety shoes while performing lathe operations while performing CNC turning operations	0.5	1	-	-
PC3. Ensure work area is clean and safe	0.5	1	-	-
PC4. Ensure that machine safety guards are in place and are in correctly working condition	0.5	1	-	-
PC5. Ensure that all tools, equipment are in safe and usable conditions	0.5	2	-	-
PC6. Ensure availability of job specification i.e. approved drawings, sketches, instructions from the supervisor, job instruction sheet/ job card	0.5	2	-	-
PC7. • Read and understand the Job requirements from the job specifications and attention shall be given • to the geometric tolerances	0.5	2	-	-
PC8. Check the work piece material for the dimensions and ensure that it is free from foreign objects, dirt or other contamination and is within the required size	0.5	2	-	-
PC9. Plan to perform the turning or other lathe operations and the sequence of operations as per required job specifications on CNC lathe machine	1	2	-	-
PC10. Obtain all the appropriate tools and measuring instruments/ gauges required for the job	1	2	-	-
<i>Operate CNC Lathe Machine</i>	8.5	20	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Check the CNC lathe machine for its functioning and ensure that it is ready for operation	1	2	-	-
PC12. Prepare the CNC lathe machine for the operations by mounting and setting the required work holding devices and cutting tools	1	2	-	-
PC13. Clarify any doubt, if any and see necessary instruction /training on the operation of the CNC Lathe machine whenever required	1	2	-	-
PC14. Hold the work piece securely and correctly, without distortion	1	2	-	-
PC15. Adjust the CNC Lathe machine settings as per job requirement to maintain desired accuracy	1	2	-	-
PC16. Perform daily maintenance of machine according to defined checklist, at the beginning of day's shifts	0.5	2	-	-
PC17. Use and extract information from engineering drawings, dimensioning and tolerances	0.5	2	-	-
PC18. Use and extract information from reference charts, tables, graphs and Engineering standards	0.5	2	-	-
PC19. Load and unload component(s) using pre-determined fixtures or work holding devices as per work instructions	1	2	-	-
PC20. Make basic program and check correctness of program through dry run and single block check	1	2	-	-
<i>Inspect and Measure Components</i>	8	27	-	-
PC21. Adjust and set the speed and feed of the CNC lathe machine to achieve the job specifications	1	2	-	-
PC22. Operate the machine tool controls safely and correctly, in line with operational procedures	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. Stop the CNC lathe machine, both in normal and emergency situations correctly by following the right procedure and should be able to restart the machine after the emergency	0.5	3	-	-
PC24. Do first part cutting trial by setting tool offsets to get oversize part	0.5	3	-	-
PC25. - Measure the critical parameters of the machined component on the machine (without removing - from the machine)	0.5	3	-	-
PC26. Correct the offsets based on the measurements by accessing program edit facility in order to enter tooling data	1	3	-	-
PC27. Measure the component after unloading to check for accuracy in the critical parameters as per job specifications	1	3	-	-
PC28. Produce machined components that combine different turning operations and have a range of features	0.5	3	-	-
PC29. Follow the specified machining sequence and procedure as per job specifications	1	3	-	-
PC30. Interpret in-built machine alarms & respond to same as per operating manual or specified instructions	1	2	-	-
<i>Maintain Machine and Workplace</i>	7.5	27	-	-
PC31. Observe for inconsistency in dimensions due to tool wear and correct the offsets accordingly	0.5	2	-	-
PC32. Ensure that machine settings are adjusted as and when required, either by self or the setter, to maintain the required accuracy	1	2	-	-
PC33. Identify when tools need replacement and replace worn tool with new tool	1	4	-	-
PC34. Produce components as per required standards	1	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC35. Report problems and seek appropriate assistance in a timely manner	0.5	4	-	-
PC36. - Complete documentation during and post operations as per organizational procedures and - applicable quality management system	1	3	-	-
PC37. Return the machine and all tools and equipment to the correct location on completion of activities	1	2	-	-
PC38. Leave the work area in a safe and tidy condition on completion of job activities as per 5S practices	0.5	2	-	-
PC39. Report the problem to the supervisor, if it cannot be resolved	0.5	2	-	-
PC40. Seek guidance from the supervisor/ specialist of the problem is outside his/her area of competence	0.5	2	-	-
NOS Total	30	90	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0705
NOS Name	CNC Lathe Machine operation
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	CNC Lathe, CNC Lathe
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0406: To Work effectively with Others

Description

This unit covers basic etiquette and competencies that an individual is required to possess and demonstrate in their behavior and interactions with others at the workplace. This covers areas such as communication, discipline, handling conflict and grievances

Scope

The scope covers the following :

- This unit/task covers the following:
- Effective Working with others

Elements and Performance Criteria

Effective Working with others

To be competent, the user/individual on the job must be able to:

- PC1.** display appropriate communication etiquette while working
- PC2.** display active listening skills while interacting with others at work
- PC3.** demonstrate responsible and disciplined behaviors at the workplace Disciplined behaviors: e.g. punctuality; completing tasks as per given time and standards; not gossiping and idling time; eliminating waste, observing honesty, etc.
- PC4.** accurately receive information & instructions from supervisor & fellow workers, getting clarification where required
- PC5.** accurately receive information & instructions from supervisor & fellow workers, getting clarification where required
- PC6.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC7.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC8.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies and procedures followed in the company for working with others in an organizational set-up
- KU2.** grievance/ conflict handling mechanism of the company
- KU3.** relevant people and their responsibilities within the work area

Generic Skills (GS)



Qualification Pack

User/individual on the job needs to know how to:

- GS1.** importance of effective communication in the workplace
- GS2.** importance of teamwork in organizational and individual success
- GS3.** barriers to effective communication
- GS4.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- GS5.** various categories of people that one is required to communicate and co- ordinate within the organization
- GS6.** importance of discipline for professional success
- GS7.** importance of ethics for professional success
- GS8.** what constitutes disciplined behavior for a working professional
- GS9.** common reasons for interpersonal conflict
- GS10.** importance and ways of managing interpersonal conflict effectively
- GS11.** importance of developing effective working relationships for professional success
- GS12.** expressing and addressing grievances appropriately and effectively

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Effective Working with others</i>	10	20	-	-
PC1. display appropriate communication etiquette while working	1	2	-	-
PC2. display active listening skills while interacting with others at work	1	2	-	-
PC3. demonstrate responsible and disciplined behaviors at the workplace Disciplined behaviors: e.g. punctuality; completing tasks as per given time and standards; not gossiping and idling time; eliminating waste, observing honesty, etc.	1	4	-	-
PC4. accurately receive information & instructions from supervisor & fellow workers, getting clarification where required	1	4	-	-
PC5. accurately receive information & instructions from supervisor & fellow workers, getting clarification where required	2	2	-	-
PC6. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	2	2	-	-
PC7. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	1	2	-	-
PC8. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	1	2	-	-
NOS Total	10	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0406
NOS Name	To Work effectively with Others
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Generic, Generic
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0408: Basics of computer and data entry

Description

This NOS is about the basics of computer and data entry like entering, updating and maintain Job work related data in the computer systems having MS Office software.

Scope

The scope covers the following :

- This unit / task covers the following
- Enter, update and maintain data in MS Office system

Elements and Performance Criteria

Enter, update and maintain data

To be competent, the user/individual on the job must be able to:

- PC1.** Fill and process mandated forms for receiving, processing, or tracking data, enter data from source documents in to Computer application having MS OFFICE software
- PC2.** Verify data entered with source documents, checks for compliance and corrects all typographical errors and missing or repeated data
- PC3.** Maintain files of source documents or other information related to data entered
- PC4.** Update database information to reflect most current source information
- PC5.** Assist in the filing and storage of security and back up data files
- PC6.** Respond to requests for information and access relevant files

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The data management applications/tools used by the company
- KU2.** Data entry protocol
- KU3.** Data integrity and security policies of the company
- KU4.** Approved methods for carrying document control and archiving
- KU5.** Approved methods for carrying document control and archiving
- KU6.** Work on MS office software

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Efficiently enter data into computer applications
- GS2.** Prepare legible reports



Qualification Pack

- GS3.** Read and understand manuals, SOPs, instructions, memos, reports, job cards etc
- GS4.** Communicate effectively with the team members and supervisors
- GS5.** Apply basic logic to identify data errors
- GS6.** Pay attention to details
- GS7.** Plan assigned tasks within timeline and as per priority order specified
- GS8.** Identify process improvements

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Enter, update and maintain data</i>	10	20	-	-
PC1. Fill and process mandated forms for receiving, processing, or tracking data, enter data from source documents in to Computer application having MS OFFICE software	2	4	-	-
PC2. Verify data entered with source documents, checks for compliance and corrects all typographical errors and missing or repeated data	2	4	-	-
PC3. Maintain files of source documents or other information related to data entered	1	4	-	-
PC4. Update database information to reflect most current source information	2	2	-	-
PC5. Assist in the filing and storage of security and back up data files	1	2	-	-
PC6. Respond to requests for information and access relevant files	2	4	-	-
NOS Total	10	20	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0408
NOS Name	Basics of computer and data entry
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Generic, Generic
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0407: Soft Skills.

Description

This NOS covers the knowledge, skills, and attitude required to communicate effectively, work as part of a team, maintain professional behavior, and handle workplace challenges confidently in a manufacturing or engineering environment

Scope

The scope covers the following :

- This NOS covers the knowledge, skills, and attitude required to communicate effectively, work as part of a team, maintain professional behavior, and handle workplace challenges confidently in a manufacturing or engineering environment

Elements and Performance Criteria

Communication and Interpersonal Skills

To be competent, the user/individual on the job must be able to:

- PC1.** • Accurately receive information and instructions from the supervisor/operator and fellow workers,
• getting clarification where required
- PC2.** • Accurately pass on information to authorized persons who require it and within agreed timescale
• and confirm its receipt
- PC3.** • Display helpful behavior by assisting others in performing tasks in a positive manner, where
• required and possible

Display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible

To be competent, the user/individual on the job must be able to:

- PC4.** • Basic Knowledge of consult with and assist others to maximize effectiveness and efficiency in
• carrying out tasks
- PC5.** Basic Study of Fundamental of Computers
- PC6.** Components of Computer: - Hardware and the software

Professionalism and Ethics

To be competent, the user/individual on the job must be able to:

- PC7.** Display active listening skills while interacting with others at work
- PC8.** Use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** Demonstrate responsible and disciplined behaviors at the workplace
- PC10.** • Escalate grievances and problems to appropriate authority as per procedure to resolve them
and
• avoid conflict

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company policies related to conduct, discipline, and communication
- KU2.** Reporting hierarchy and escalation procedures
- KU3.** Importance of teamwork and coordination in achieving production goals
- KU4.** Documentation and reporting standards followed in the organization
- KU5.** Importance of maintaining professionalism to uphold the company reputation
- KU6.** Principles of effective verbal and written communication
- KU7.** Listening, questioning, and feedback techniques
- KU8.** Types of non-verbal communication and their significance
- KU9.** Common workplace challenges and appropriate problem-solving approaches
- KU10.** Basics of time management, planning, and prioritization
- KU11.** Importance of ethics, integrity, and respect at work
- KU12.** Emotional intelligence recognizing and managing one own emotions
- KU13.** Conflict resolution techniques and stress management practices
- KU14.** Health, hygiene, and safety practices in an industrial setup
- KU15.** Health, hygiene, and safety practices in an industrial setup

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Speak clearly, confidently, and politely in meetings or discussions
- GS2.** Write simple, grammatically correct notes, emails, and forms
- GS3.** Listen actively and interpret messages correctly
- GS4.** Provide feedback constructively without offending others
- GS5.** Communicate across hierarchical levels respectfully
- GS6.** Identify causes of common workplace issues
- GS7.** Think critically and propose practical solutions
- GS8.** Make sound decisions within scope of authority
- GS9.** Use data or feedback to improve performance
- GS10.** Anticipate potential problems and plan preventive actions
- GS11.** Build positive relationships with team members
- GS12.** Respect cultural, gender, and generational differences
- GS13.** Collaborate effectively to complete joint tasks
- GS14.** Motivate peers and maintain team morale
- GS15.** Handle criticism and feedback maturely
- GS16.** Manage emotions and stress during high-pressure situations
- GS17.** Demonstrate self-confidence and a learning attitude



Qualification Pack

- GS18.** Seek feedback to improve own performance
- GS19.** Stay open to new technologies, methods, and processes
- GS20.** Exhibit adaptability to workplace or role changes

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communication and Interpersonal Skills</i>	3	6	-	-
PC1. • Accurately receive information and instructions from the supervisor/operator and fellow workers, • getting clarification where required	1	2	-	-
PC2. • Accurately pass on information to authorized persons who require it and within agreed timescale • and confirm its receipt	1	2	-	-
PC3. • Display helpful behavior by assisting others in performing tasks in a positive manner, where • required and possible	1	2	-	-
<i>Display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible</i>	3	6	-	-
PC4. • Basic Knowledge of consult with and assist others to maximize effectiveness and efficiency in • carrying out tasks	1	2	-	-
PC5. Basic Study of Fundamental of Computers	1	2	-	-
PC6. Components of Computer: - Hardware and the software	1	2	-	-
<i>Professionalism and Ethics</i>	4	8	-	-
PC7. Display active listening skills while interacting with others at work	1	2	-	-
PC8. Use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	1	2	-	-
PC9. Demonstrate responsible and disciplined behaviors at the workplace	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. - Escalate grievances and problems to appropriate authority as per procedure to resolve them and - avoid conflict	1	2	-	-
NOS Total	10	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0407
NOS Name	Soft Skills.
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Generic, Generic
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0401: Basic health and safety practices at the workplace, 5S

Description

This unit is about creating a Safe and Healthy work place, adhering to the safety guidelines in the working area, following practices which are not impacting the environment in a negative manner. This NOS is about ensuring all 5S activities both at the shop floor and the office area to facilitate increase in work productivity. It covers knowledge of fire safety, common first aid applications, safe practice and emergency applications.

Scope

The scope covers the following :

- The role holder will be responsible for
- Identifying and reporting of risks
- Creating and sustaining a safe, clean and environment friendly Work place.
- Ensure sorting, streamlining & organizing, storage and documentation, cleaning, standardization and sustenance across the plant and office premises of the organization

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** Wear protective clothing/equipment for specific tasks and work conditions
- PC2.** Carry out safe working practices while dealing with hazards to ensure the safety of Self and others.
- PC3.** Apply good housekeeping practices at all times

Fire safety

To be competent, the user/individual on the job must be able to:

- PC4.** Use the various appropriate fire extinguishers on different types of fires correctly
- PC5.** Demonstrate rescue techniques applied during fire hazard, demonstrate good housekeeping in order to prevent fire hazards, demonstrate the correct use of a fire extinguisher.

Identify and report the risks identified

To be competent, the user/individual on the job must be able to:

- PC6.** Identify activities which can cause potential injury through sharp objects, burns, fall, electricity, gas leakages, radiation, poisonous fumes, chemicals, loud noise, and Identify areas in the plant which are potentially hazardous/ unhygienic in nature. Conduct regular checks with support of the maintenance team on machine health to identify potential hazards due to wear and tear of machine.
- PC7.** Inform the concerned authorities on the potential risks identified in the processes, workplace area/ layout, materials used etc, Inform the concerned authorities about machine breakdowns, damages which can potentially harm man/ machine during operations.
- PC8.** Create awareness amongst other by sharing information on the identified risks.

Ensure sorting

Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC9.** Follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.
- PC10.** Ensure segregation of waste in hazardous/ non Hazardous waste as per the sorting work instructions
- PC11.** Follow the technique of waste disposal and waste storage in the proper bins as per SOP
- PC12.** Segregate the items which are labeled as red tag items for the process area and keep them in the correct places
- PC13.** Sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/ work instructions
- PC14.** Ensure that areas of material storage areas are not overflowing
- PC15.** Properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required
- PC16.** Return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area
- PC17.** Follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards

Ensure proper documentation and storage(organizing, streamlining)

To be competent, the user/individual on the job must be able to:

- PC18.** Follow the proper labelling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists
- PC19.** Check that the items in the respective areas have been identified as broken or damaged
- PC20.** Follow the given instructions and check for labelling of fluids, oils, lubricants, solvents, chemicals etc. and proper storage of the same. To avoid spillage, leakage, fire etc.
- PC21.** Make sure that all material and tools are stored in the designated places and in the manner indicated in the 5S instructions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Relevant standards, procedures and policies related to Health, Safety and Environment followed in the company
- KU2.** Emergency handling procedures & hierarchy for escalation
- KU3.** Basic knowledge of Safety procedures(firefighting, first aid) within the organization
- KU4.** Basic knowledge of various types of PPEs and their usage
- KU5.** Basic knowledge of risks/hazards associated with each occupation in the organization
- KU6.** Knowledge of personal hygiene and how an individual can contribute towards creating a highly safe and clean working environment
- KU7.** Meaning of hazards and risks
- KU8.** Health and safety hazards commonly present in the work environment and related precautions

Qualification Pack

- KU9.** Possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
- KU10.** Possible causes of risk and accident (due to oil leakage)
- KU11.** Safe working practices when working with tools and machines
- KU12.** Safe working practices while working at various hazardous sites
- KU13.** Where to find all the general health and safety equipment in the workplace
- KU14.** Various dangers associated with the use of electrical equipment
- KU15.** Preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU16.** Importance of using protective clothing/equipment while working
- KU17.** Precautionary activities to prevent the fire accident
- KU18.** Various causes of fire
- KU19.** Techniques of using the different fire extinguishers
- KU20.** Different methods of extinguishing fire
- KU21.** Different materials used for extinguishing fire
- KU22.** Rescue techniques applied during a fire hazard
- KU23.** Various types of safety signs and what they mean
- KU24.** Appropriate basic first aid treatment relevant to the condition eg. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
- KU25.** Content of written accident report
- KU26.** Potential injuries and ill health associated with incorrect manual handling
- KU27.** Safe lifting and carrying practices
- KU28.** Personal safety, health and dignity issues relating to the movement of a person by others
- KU29.** Potential impact to a person who is moved incorrectly
- KU30.** Have basic knowledge of 5S procedures
- KU31.** Know various types 5s practices followed in various areas
- KU32.** Understand the 5S checklists provided in the department/ team
- KU33.** Have skills to identify useful & non useful items
- KU34.** Have knowledge of labels , signs & colors used as indicators
- KU35.** Have knowledge on how to sort and store various types of tools, equipment, material etc.
- KU36.** Know , how to identify various types of waste products
- KU37.** Understand the impact of waste/ dirt/ dust/unwanted substances on the process/ environment/ machinery/ human body.
- KU38.** Have knowledge of best ways of cleaning & waste disposal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Write basic level notes and observations
- GS2.** Read safety instructions put up across the plant premises

Qualification Pack

- GS3.** Read safety precautions mentioned in equipment manuals and panels to understand the potential risks associated
- GS4.** Effectively communicate information to team members
- GS5.** Inform employees in the plant and concerned functions about events, incidents & potential risks observed related to Safety, Health and Environment.
- GS6.** Question operator/ supervisor in order to understand the safety related issues
- GS7.** Attentively listen with full attention and comprehend the information given by the speaker during safety drills and training programs
- GS8.** Plan and organize the work order and jobs received from the internal customers
- GS9.** Plan and organize the design documents received from internal customers
- GS10.** Organize all process/ equipment manuals so that sorting out information is Fast
- GS11.** Use common sense and make judgments during day to day basis
- GS12.** Use reasoning skills to identify and resolve basic problems
- GS13.** Use intuition to detect any potential problems which could arise during operations
- GS14.** Follow instructions and work on areas of improvement identified
- GS15.** Complete the assigned tasks with minimum supervision
- GS16.** Complete the job defined by the supervisor within the timelines and quality norms

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	1.5	6	-	-
PC1. Wear protective clothing/equipment for specific tasks and work conditions	0.5	2	-	-
PC2. Carry out safe working practices while dealing with hazards to ensure the safety of Self and others.	0.5	2	-	-
PC3. Apply good housekeeping practices at all times	0.5	2	-	-
<i>Fire safety</i>	1	4	-	-
PC4. Use the various appropriate fire extinguishers on different types of fires correctly	0.5	2	-	-
PC5. Demonstrate rescue techniques applied during fire hazard, demonstrate good housekeeping in order to prevent fire hazards, demonstrate the correct use of a fire extinguisher.	0.5	2	-	-
<i>Identify and report the risks identified</i>	1.5	6	-	-
PC6. Identify activities which can cause potential injury through sharp objects, burns, fall, electricity, gas leakages, radiation, poisonous fumes, chemicals, loud noise, and Identify areas in the plant which are potentially hazardous/ unhygienic in nature. Conduct regular checks with support of the maintenance team on machine health to identify potential hazards due to wear and tear of machine.	0.5	2	-	-
PC7. Inform the concerned authorities on the potential risks identified in the processes, workplace area/ layout, materials used etc, Inform the concerned authorities about machine breakdowns, damages which can potentially harm man/ machine during operations.	0.5	2	-	-
PC8. Create awareness amongst other by sharing information on the identified risks.	0.5	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure sorting</i>	4	10	-	-
PC9. Follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.	0.5	2	-	-
PC10. Ensure segregation of waste in hazardous/ non Hazardous waste as per the sorting work instructions	0.5	2	-	-
PC11. Follow the technique of waste disposal and waste storage in the proper bins as per SOP	0.5	1	-	-
PC12. Segregate the items which are labeled as red tag items for the process area and keep them in the correct places	0.5	1	-	-
PC13. Sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/ work instructions	0.5	1	-	-
PC14. Ensure that areas of material storage areas are not overflowing	0.25	0.5	-	-
PC15. Properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required	0.25	0.5	-	-
PC16. Return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area	0.5	1	-	-
PC17. Follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards	0.5	1	-	-
<i>Ensure proper documentation and storage(organizing, streamlining)</i>	2	4	-	-
PC18. Follow the proper labelling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists	0.5	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. Check that the items in the respective areas have been identified as broken or damaged	0.5	1	-	-
PC20. Follow the given instructions and check for labelling of fluids, oils, lubricants, solvents, chemicals etc. and proper storage of the same. To avoid spillage, leakage, fire etc.	0.5	1	-	-
PC21. Make sure that all material and tools are stored in the designated places and in the manner indicated in the 5S instructions	0.5	1	-	-
NOS Total	10	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0401
NOS Name	Basic health and safety practices at the workplace, 5S
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Generic
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

Qualification Pack

PC26. identify different types of customers

PC27. identify and respond to customer requests and needs in a professional manner.

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:



Qualification Pack

- GS1.** read and write different types of documents/instructions/correspondence
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Career Development & Goal Setting</i>	1	2	-	-
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIP or email
- Assessment agencies send the assessment confirmation to VTP/TC looping SSC
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- SSC monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 40, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

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3. Assessment Quality Assurance levels/Framework:

- Question bank is created by the Subject Matter Experts (SME) are verified by the other SME
- Questions are mapped to the specified assessment criteria
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geo tagged reporting of the assessor from assessment location
- Centre photographs with signboards and scheme specific branding

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Method for assessment documentation, archiving, and access

- Hard copies of the documents are stored

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CPC/CAP/N0701.Basic Engineering	30	50	-	-	80	15
CPC/CAP/N0702.Metal Cutting & Cutting Tools	20	20	-	-	40	15
CPC/CAP/N0703.Conventional Lathe Machines	30	60	-	-	90	15
CPC/CAP/N0704.CNC Lathe Machine Programming	30	80	-	-	110	15

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CPC/CAP/N0705.CNC Lathe Machine operation	30	90	-	-	120	15
CPC/CAP/N0406.To Work effectively with Others	10	20	-	-	30	5
CPC/CAP/N0408.Basics of computer and data entry	10	20	-	-	30	5
CPC/CAP/N0407.Soft Skills.	10	20	-	-	30	5
CPC/CAP/N0401.Basic health and safety practices at the workplace, 5S	10	30	-	-	40	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	200	420	-	-	620	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.